Rhysida Ransomware Simulation

23CSE334 – Cyber Forensics and Malware

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Threat Model

Assumptions & Surface

- Windows enterprise endpoint.
- Outbound HTTPS allowed.
- User can execute attachments.
- Attack surface: Startup folder, HKCU registry, PowerShell, user directories.

Attacker Objectives

- **Initial Access:** Phishing emails.
- **Privilege Escalation:** UAC bypass.
- **Persistence:** Startup folder abuse.
- **Payload:** Cloud-hosted delivery.
- **Encryption:** Deny availability.
- **Anti-Forensics:** Artifact deletion.

Methodology: Attack Lifecycle

Based on MITRE ATT&CK Framework:

1. **Reconnaissance:** Gathering intelligence on weak credentials/services.
2. **Delivery:** Phishing email attachment triggers the loader.
3. **Exploitation:** Using PowerShell to escalate privileges.
4. **Installation & C2:** Persistence via Startup folder; payload downloaded via HTTPS.
5. **Actions on Objectives:** Ransomware deployment, file encryption (.soma), and ransom note generation.

Forensic Toolkit

Tool	Purpose
Oracle VirtualBox	Isolated ransomware execution environment
Guymager	Forensic disk image acquisition
Autopsy	File system and deleted file analysis
DumpIt & Volatility	Volatile memory acquisition and analysis
Wireshark	Network traffic analysis
Autoruns	Persistence detection
Sigma Rules	Behavioral detection
VirusTotal	Threat intelligence analysis

Implementation: Delivery & Execution

- **Delivery:** Phishing email with a disguised malicious document.
- **Execution:** Attachment launches a batch script invoking hidden PowerShell.
- **Evasion:**
  - PowerShell downloads payloads from a cloud repository over HTTPS.
  - Blends with legitimate cloud traffic to bypass perimeter security.

Evidence: Memory Forensics

Technique: **Volatile Memory Capture**

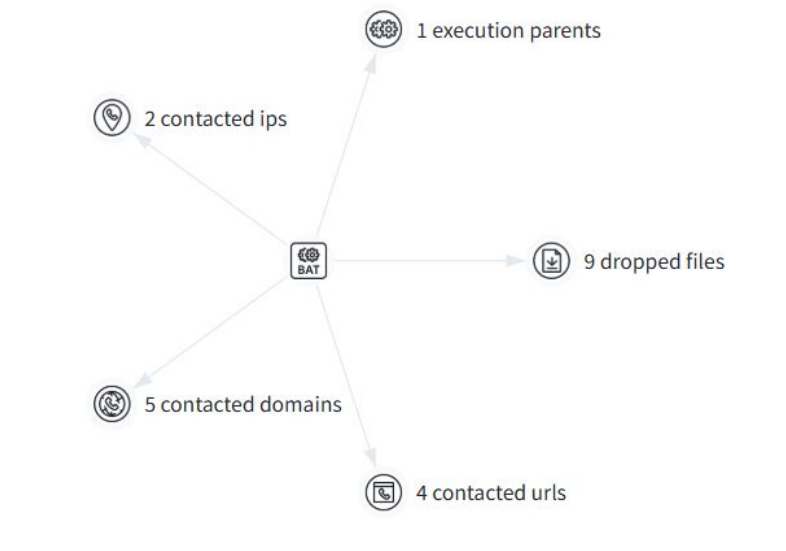
Memory acquisition preserves transient artifacts that exist only in RAM, including:

- Running ransomware processes.
- Malicious command-line arguments.
- Injected scripts bypassing disk storage.

6124	6004	cmd.exe	0xcd0fb56f0080	3	-	1	False	2026-03-01 09:18:39.000000 UTC	N/A	Disabled
1968	6124	conhost.exe	0xcd0fb56f7080	6	-	1	False	2026-03-01 09:18:39.000000 UTC	N/A	Disabled
2336	6124	powershell.exe	0xcd0fb3fca2c0	24	-	1	False	2026-03-01 09:18:40.000000 UTC	N/A	Disabled

Evidence: Sandbox Analysis

Technique: Sandboxing



Security vendors' analysis Do

ALYac	Generic.UAC.Marte.B.61A93F4F	Arcabit	Generic.UAC.Marte.B.61A93F4F
BitDefender	Generic.UAC.Marte.B.61A93F4F	CTX	Batch.unknown.marte
Emsisoft	Generic.UAC.Marte.B.61A93F4F (B)	eScan	Generic.UAC.Marte.B.61A93F4F
GData	Generic.UAC.Marte.B.61A93F4F	Microsoft	Trojan:Script/Wacatac.B!ml
VIPRE	Generic.UAC.Marte.B.61A93F4F	Acronis (Static ML)	Undetected

Evidence: Process Analysis

Technique: **Process Tree Tracing**

Process analysis reveals a chained execution model (cmd.exe → powershell.exe).

This *Living-off-the-Land* (LotL) behavior allows the ransomware to evade traditional signature-based AV detection.

Process and service actions

Processes Created

- "C:\Windows\system32\cmd.exe" /c start /wait "" "C:\Users\<USER>\Desktop\loader.bat"
- C:\Windows\System32\svchost.exe -k LocalSystemNetworkRestricted -p -s StorSvc
- C:\Windows\System32\svchost.exe -k NetworkService -p
- C:\Windows\system32\cmd.exe /K "C:\Users\<USER>\Desktop\loader.bat"
- C:\Windows\system32\cmd.exe /K "C:\Users\<USER>\Documents\encoder.bat"
- C:\Windows\system32\lsass.exe
- C:\Windows\system32\net1 session
- C:\Windows\system32\services.exe
- C:\Windows\system32\svchost.exe -k LocalService -s W32Time
- C:\Windows\system32\svchost.exe -k UnistackSvcGroup

Evidence: File System Artifacts

Technique: Disk Forensics

Analyzing the disk image identifies newly created malicious artifacts, enabling timeline reconstruction:

- Encrypted user files (.soma).
- Ransom notes.
- Dropped persistence scripts.

Files Written

C:\Users\<USER>\AppData\Local\Microsoft\CLR_v4.0_32\UsageLogs\powershell.exe.log

C:\Users\<USER>\AppData\Local\Microsoft\Windows\PowerShell\StartupProfileData-NonInteractive

C:\Users\<USER>\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\sys_service.bat

C:\Users\<USER>\Desktop\Documents.lnk.soma

C:\Users\<USER>\Desktop\decoder.bat

C:\Users\<USER>\Desktop\soma.txt

C:\Users\<USER>\Documents\encoder.bat

+ C:\Windows\ServiceProfiles\LocalService\AppData\Local\FontCache\Fonts\Download-1.tmp

+ C:\Users\user\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\sys_service.bat

+ C:\Users\user\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\sys_service.bat\:Zone.Identifier:$DATA

Evidence: Deleted Files Analysis

Technique: Deleted File Recovery

The ransomware attempts to cover its tracks by deleting initial loader components.

Successful recovery of these files confirms deliberate anti-forensic behavior and provides the original attack payload.

Dropped Files (9)

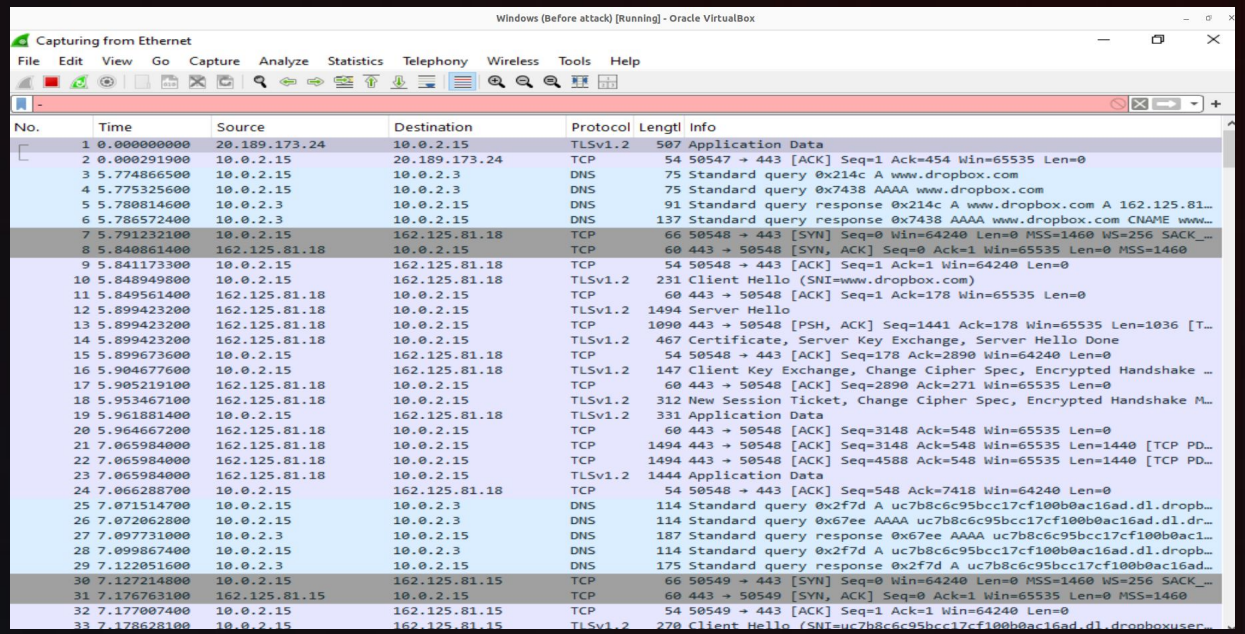
Scanned	Detections	File type	Name
2026-03-01	0 / 62	DOS batch file	decoder.bat
2026-02-16	0 / 62	INI	MasonClient.exe:Zone.Identifier
2026-03-01	0 / 62	Text	soma.txt
2026-03-01	9 / 62	DOS batch file	loader.bat
2026-02-27	0 / 62	Powershell	__PSScriptPolicyTest_4kkpskmn.3fe.psm1
2026-03-01	0 / 61	DOS batch file	encoder.bat
2026-02-18	0 / 62	Text	powershell.exe.log
2026-02-15	0 / 62	Text	Null
?	?	file	da481edc5572abdb723c2d7513006f001d753c930d7059816d268e63aa6b3cc4

Evidence: Network Traffic

Technique: **Packet & Domain Analysis**

Network logs capture outbound HTTPS connections to legitimate cloud infrastructure.

This verifies remote payload delivery and Command & Control (C2) communication masking as normal traffic.



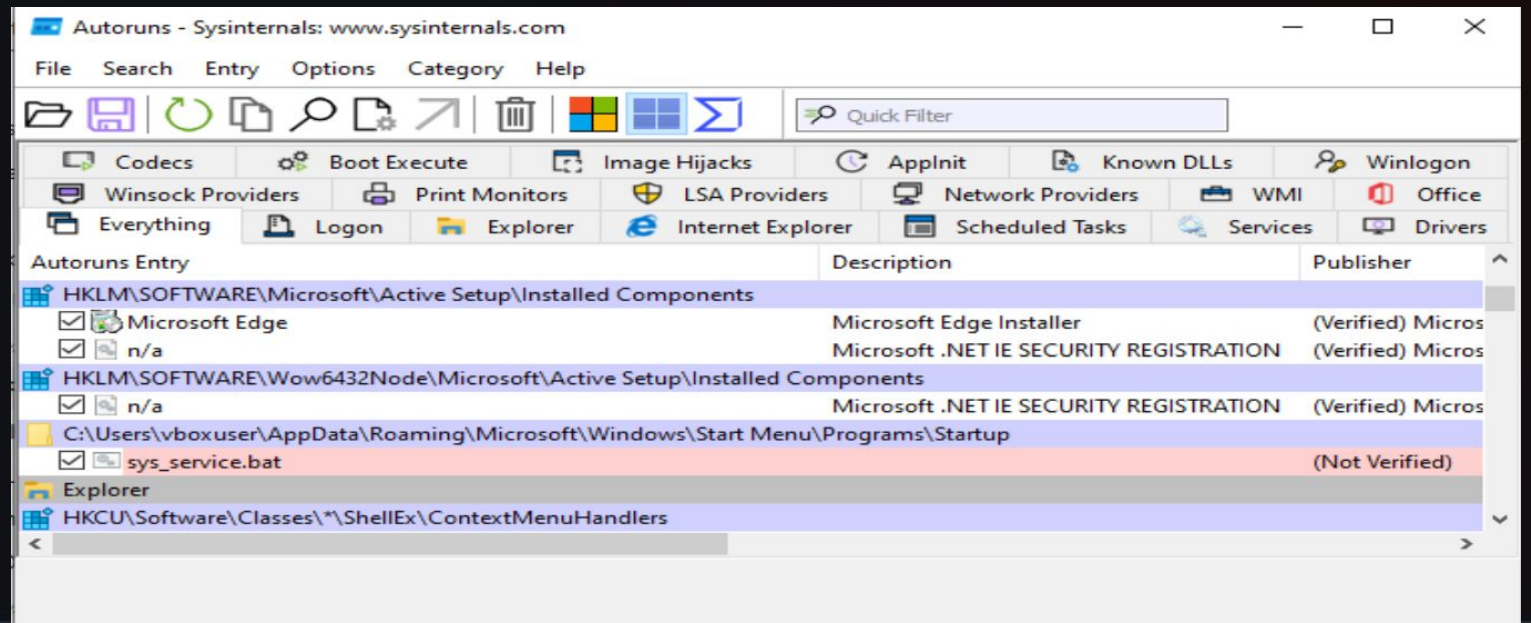
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	20.189.173.24	10.0.2.15	TLSv1.2	507	Application Data
2	0.000291900	10.0.2.15	20.189.173.24	TCP	54	50547 → 443 [ACK] Seq=1 Ack=454 Win=65535 Len=0
3	5.774866500	10.0.2.15	10.0.2.3	DNS	75	Standard query 0x214c A www.dropbox.com
4	5.775325600	10.0.2.15	10.0.2.3	DNS	75	Standard query 0x7438 AAAA www.dropbox.com
5	5.780814600	10.0.2.3	10.0.2.15	DNS	91	Standard query response 0x214c A www.dropbox.com A 162.125.81…
6	5.786572400	10.0.2.3	10.0.2.15	DNS	137	Standard query response 0x7438 AAAA www.dropbox.com CNAME www…
7	5.791232100	10.0.2.15	162.125.81.18	TCP	66	50548 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_…
8	5.840861400	162.125.81.18	10.0.2.15	TCP	60	443 → 50548 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
9	5.841173300	10.0.2.15	162.125.81.18	TCP	54	50548 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
10	5.848949800	10.0.2.15	162.125.81.18	TLSv1.2	231	Client Hello (SNI=www.dropbox.com)
11	5.849561400	162.125.81.18	10.0.2.15	TCP	60	443 → 50548 [ACK] Seq=1 Ack=178 Win=65535 Len=0
12	5.899423200	162.125.81.18	10.0.2.15	TLSv1.2	1494	Server Hello
13	5.899423200	162.125.81.18	10.0.2.15	TCP	1090	443 → 50548 [PSH, ACK] Seq=1441 Ack=178 Win=65535 Len=1036 [T…
14	5.899423200	162.125.81.18	10.0.2.15	TLSv1.2	467	Certificate, Server Key Exchange, Server Hello Done
15	5.899673600	10.0.2.15	162.125.81.18	TCP	54	50548 → 443 [ACK] Seq=178 Ack=2890 Win=64240 Len=0
16	5.904677600	10.0.2.15	162.125.81.18	TLSv1.2	147	Client Key Exchange, Change Cipher Spec, Encrypted Handshake …
17	5.905219100	162.125.81.18	10.0.2.15	TCP	60	443 → 50548 [ACK] Seq=2890 Ack=271 Win=65535 Len=0
18	5.953467100	162.125.81.18	10.0.2.15	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted Handshake M…
19	5.961881400	10.0.2.15	162.125.81.18	TLSv1.2	331	Application Data
20	5.964667200	162.125.81.18	10.0.2.15	TCP	60	443 → 50548 [ACK] Seq=3148 Ack=548 Win=65535 Len=0
21	7.065984000	162.125.81.18	10.0.2.15	TCP	1494	443 → 50548 [ACK] Seq=3148 Ack=548 Win=65535 Len=1440 [TCP PD…
22	7.065984000	162.125.81.18	10.0.2.15	TCP	1494	443 → 50548 [ACK] Seq=4588 Ack=548 Win=65535 Len=1440 [TCP PD…
23	7.065984000	162.125.81.18	10.0.2.15	TLSv1.2	1444	Application Data
24	7.066288700	10.0.2.15	162.125.81.18	TCP	54	50548 → 443 [ACK] Seq=548 Ack=7418 Win=64240 Len=0
25	7.071514700	10.0.2.15	10.0.2.3	DNS	114	Standard query 0x2f7d A uc7b8c6c95bcc17cf100b0ac16ad.dl.dropb…
26	7.072062800	10.0.2.15	10.0.2.3	DNS	114	Standard query 0x67ee AAAA uc7b8c6c95bcc17cf100b0ac16ad.dl.dr…
27	7.097731000	10.0.2.3	10.0.2.15	DNS	187	Standard query response 0x67ee AAAA uc7b8c6c95bcc17cf100b0ac1…
28	7.099867400	10.0.2.15	10.0.2.3	DNS	114	Standard query 0x2f7d A uc7b8c6c95bcc17cf100b0ac16ad.dl.dropb…
29	7.122051600	10.0.2.3	10.0.2.15	DNS	175	Standard query response 0x2f7d A uc7b8c6c95bcc17cf100b0ac16ad…
30	7.127214800	10.0.2.15	162.125.81.15	TCP	66	50549 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_…
31	7.176763100	162.125.81.15	10.0.2.15	TCP	60	443 → 50549 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460
32	7.177007400	10.0.2.15	162.125.81.15	TCP	54	50549 → 443 [ACK] Seq=1 Ack=1 Win=64240 Len=0
33	7.178628100	10.0.2.15	162.125.81.15	TLSv1.2	270	Client Hello (SNI=uc7b8c6c95bcc17cf100b0ac16ad.dl.dropboxuser…

Evidence: Persistence Mechanisms

Technique: **Autoruns Verification**

Attackers ensure the ransomware executes even after a system reboot.

Inspection reveals malicious scripts embedded directly into the Windows Startup folder, guaranteeing continuous execution.



Evidence: Framework Mapping

Technique: MITRE ATT&CK Alignment

TA0002	4 Techniques	TA0003	3 Techniques	TA0004	3 Techniques	TA0005	5 Techniques
Command and Scriptin…	4	Modify Registry	2	Process Injection	1	Masquerading	1
T1059		T1112		T1055		T1036	
Scripting	5	Boot or Logon Autostar…	1	Boot or Logon Autostar…	1	Process Injection	1
T1064		T1547		T1547		T1055	
Native API	1					Scripting	5
T1106						T1064	
						Modify Registry	2
						T1112	
						Indirect Command Execution	2
						T1202	

Enterprise Defense Strategy

A Layered Defense Approach:

- **Endpoint Controls:** Constrained Language Mode for PowerShell, enhanced logging (Sysmon), application whitelisting, and strict startup monitoring.
- **Network Controls:** Egress filtering, DNS inspection, and TLS monitoring to detect anomalous cloud communications.
- **Resilience:** Offline, immutable backups to negate extortion leverage.
- **Incident Response:** Forensic readiness and tailored playbooks for rapid containment.

Conclusion

- Successfully reconstructed the **Rhysida attack lifecycle** using multi-layered forensic analysis.
- Demonstrated how modern ransomware leverages trusted tools and cloud infrastructure for stealth.
- Highlighted that defending against such threats requires defense-in-depth, strict execution controls, and robust forensic readiness.